Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs

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Abstract

Under the Position-Based Model (PBM), the probability that a user clicks a document d shown for query q at position k factorises as $p_k\, r_{q,d}$, where p_k is the position examination probability and $r_{q,d}$ is the query–document relevance. Intervention harvesting [1] estimates the ratio $p_k/p_{k'}$ from naturally occurring ranking variation by aggregating per-cell click-rate ratios over (q,d) cells. On real logs, cell impression counts are extremely heavy-tailed: on the full KDD Cup 2012 Track 2 log, 60% of interventional pairs have at most five impressions while 0.1% of pairs account for 42% of all impressions. The weighting of cells is therefore central to finite-sample efficiency.

We study count-only weighting schemes for intervention harvesting. Beyond the uniform weighting of Agarwal et al., we propose two new count-only weights: the min-count weight $\omega_i = \min(N_k^i, N_{k'}^i)$ and the harmonic weight $\omega_i = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$, where N_k^i is the number of impressions of cell i at position k. The harmonic weight arises from an inverse-variance approximation to per-cell click-rate differences under PBM: after replacing the relevance-dependent variance terms with a common factor, the resulting count-only *surrogate variance* is minimised by harmonic weighting. A Cauchy–Schwarz argument shows that harmonic improves this surrogate variance over uniform aggregation for any impression counts.

We evaluate finite-sample efficiency via the closed-form delta-method asymptotic variance of the weighted ratio estimator [10, 12]. On the full KDD log, harmonic weighting achieves a 4.5× asymptotic relative efficiency improvement over uniform and remains robust under weight capping. Synthetic and Yahoo-LTR semi-synthetic experiments with ground truth corroborate the prediction, with up to 54% variance reduction under high imbalance. We also make explicit an estimand caveat: under PBM mis-specification, different weights target different weighted averages of cell-specific ratios. Overall, harmonic weighting offers a simple count-only replacement for uniform intervention harvesting with a count-only surrogate-variance guarantee and consistent empirical gains; min and harmonic perform comparably, with harmonic preferred for its theoretical justification.

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CCS Concepts

• **Information systems → Learning to rank**; *Evaluation of retrieval results*; Web search engines.

Keywords

position bias, intervention harvesting, unbiased learning to rank, propensity estimation, click models, inverse-variance weighting

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1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [7]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k/p_{k'}$ from naturally occurring ranking variations without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions $(1,2)$), **60.7% of pairs have $N_1 + N_2 \le 5$ while 0.1% of pairs carry 42% of all impressions** [8]. Public benchmarks can hide this shape: the released Baidu-ULTR [15] aggregation has 94% of (q,d,pos) cells with $N = 1$ and 78% of adjacent-pair impression ratios within $[0.5, 2]$, which we attribute to the per-cell deduplication used in the public release rather than to the underlying production logs (the unaggregated KDD release, by contrast, preserves the heavy-tailed production distribution).

Equal-weighting heteroscedastic estimators is statistically suboptimal by classical inverse-variance theory [4], a result the meta-analysis literature has championed for decades. The intervention-harvesting estimator of [1] uses uniform per-cell weighting (each cell's click rate enters the sum with weight 1). **We propose two new count-aware alternatives**: the *min-count* weight $\omega_i = \min(N_k^i, N_{k'}^i)$ and the *harmonic* weight $\omega_i = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$. Empirically, min and harmonic perform comparably on KDD (Section 4); the harmonic proposal is preferable because of its stronger theoretical foundation: it is the inverse-variance optimum of a relevance-agnostic surrogate (Theorem 2) and the only one of the two with an unconditional variance-reduction guarantee against uniform (Theorem 3). Existing variance-reduction arguments on heavy-tailed real logs have relied on *bootstrap* resampling, which we show can be misleading for inverse-variance-style weights on singleton-dominated data (Section 3).

Contributions. Inverse-variance weighting is classical; our contribution is not the principle itself but its adaptation to, and diagnosis on, heavy-tailed intervention harvesting. **(1)** We show the harmonic weight is the inverse-variance optimum of a *relevance-agnostic count-only surrogate* of the per-cell variance under PBM (Theorem 2); dropping the relevance-dependent term is precisely what makes the optimum *deployable*, since the exact oracle needs per-cell relevances. We prove a Cauchy–Schwarz guarantee that harmonic reduces this surrogate for any impression counts (Theorem 3) and characterise the exact asymmetric oracle, of which harmonic is the parameter-free case (Theorem 4). **(2)** We report a counter-intuitive *oracle collapse*: the exact plug-in inverse-variance oracle, which uses *estimated* per-cell relevances, is *worse* than uniform on heavy-tailed logs (ARE 0.002, Table 2), because $\hat{r}_i$ is unusable on the ~60% singleton mass – so the count-only surrogate is not a convenience but the robust choice. **(3)** We show that naive pair-bootstrap is the wrong efficiency diagnostic on singleton-dominated data, and adopt the closed-form asymptotic variance of the weighted ratio estimator, properly aligned with conditional-on-counts efficiency. **(4)** On the full KDD log, harmonic delivers an asymptotic relative efficiency of 4.53 over uniform – uniform would need ≈4.5× the data to match harmonic's precision – with the plug-in within 1–2% of empirical click-resample standard error and robust to weight capping. A Cochran-Q heterogeneity test (Q/df = 3.83) confirms substantial PBM mis-specification, under which the choice of weight is a precision/estimand trade-off rather than a search for a single "true" position-bias ratio. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

2 Estimator and Theory

Notation. Throughout, q is a query, d a document/ad, k a result position, and i indexes the (q,d) cells in the interventional set. The Position-Based Model (PBM) assumes $P(\text{click} \mid q,d,k) = p_k r_{q,d}$ with $p_k \in (0,1]$ the position-k examination probability and $r_{q,d} \in [0,1]$ the (q,d) relevance; we abbreviate the per-cell relevance as r_i and write the true per-cell click probability as $\theta^i_k = p_k r_i$. We observe N^i_k impressions and C^i_k clicks for cell i at position k, with empirical rate $\hat{\theta}^i_k = C^i_k/N^i_k$. Our target is the position-bias ratio $R_{k,k'} = p_k/p_{k'}$. Per-cell weights are $\omega_i \geq 0$. Given the set of (q,d) pairs that appeared at both positions k and k', the estimator of [1] for $p_k/p_{k'}$ is

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \hat{\theta}^i_k}{\sum_i \omega_i \hat{\theta}^i_{k'}}, \quad \hat{\theta}^i_k = \frac{C^i_k}{N^i_k}. \tag{1}$$

$\hat{R}_{k,k'}$ is the standard *self-normalised weighted ratio* estimator: a sum of weighted per-cell click rates divided by another such sum. We say a weight scheme is *ratio-unbiased* when, under PBM, the ratio of the conditional expectations of the numerator and denominator equals $p_k/p_{k'}$ – note this is not the same as $\mathbb{E}[\hat{R}] = p_k/p_{k'}$, because expectation does not commute with ratios; the finite-sample bias of $\hat{R}$ is $O(1/N)$ under standard regularity. Under PBM, every count-only weight $\omega_i = \omega(N^i_k, N^i_{k'})$ satisfies this property, and $\hat{R}$ is consistent for $p_k/p_{k'}$ [1].

Proposition 1 (Conditional delta-method variance). *Given the vector of impression counts $\mathbf{N} = \{N^i_k\}$, the delta method [12] gives*

$$\mathrm{Var}(\hat{R}_{k,k'} \mid \mathbf{N}) \;\approx\; R^2_{k,k'}\left[\alpha\, V_k(\omega) + \beta\, V_{k'}(\omega)\right],$$

with $V_r(\omega) = \sum_i \omega_i^2/N^i_r/(\sum_i \omega_i)^2$, $\alpha = (1-p_k)/p_k$, and $\beta = (1-p_{k'})/p_{k'}$.

Relevance-agnostic count-only IVW.. The exact Bernoulli variance is $\mathrm{Var}(\hat{\theta}^i_k) = p_k r_i(1-p_k r_i)/N^i_k$, which depends on the cell relevance r_i. Plug-in inverse-variance weights with $\hat{r}_i$ estimated per cell define the *θ-aware oracle* of Theorem 4; in heavy-tailed click logs $\hat{r}_i$ is itself extremely noisy on the ~60% singleton mass, making this oracle infeasible in practice (Table 2: ARE = 0.002). We therefore work with a *relevance-agnostic surrogate* that drops the r_i-dependent factor from the cell-level variance; this surrogate is exact when the per-cell relevance r_i is constant across cells, and a controlled approximation otherwise. The inverse-variance optimum [4] on this surrogate, restricted to count-only weights, is the harmonic mean of the two cell counts. We make this precise:

Theorem 2 (Count-only IVW on the surrogate). *Among all non-negative weights $\omega_i = \omega(N^i_k, N^i_{k'})$, harmonic weighting minimises the count-only delta-method variance surrogate $V_k(\omega) + V_{k'}(\omega)$.*

Theorem 3 (Surrogate guarantee, unconditional in counts). *For any impression counts $\{N^i_k, N^i_{k'}\}_{i=1}^n$, $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(\mathbf{1}) + V_{k'}(\mathbf{1})$. The guarantee is unconditional in the counts but conditional on $V_k + V_{k'}$ as the objective; it implies improved $\mathrm{Var}(\hat{R} \mid \mathbf{N})$ via Proposition 1 when $\alpha \approx \beta$. For the asymmetric $\alpha V_k + \beta V_{k'}$ surrogate the bound is empirical only, but we verify on full KDD that harmonic reduces it by essentially the same factor as the symmetric one (Section 4).*

Sketch. Substituting $\omega^h_i = N^i_k N^i_{k'}/(N^i_k + N^i_{k'})$ collapses the LHS to $1/W$ with $W = \sum_i \omega^h_i$. The RHS is $(1/n^2)\sum_i 1/\omega^h_i$. Cauchy–Schwarz gives $n^2 \leq W \sum_i 1/\omega^h_i$. □

Theorem 4 (Exact asymmetric oracle). *$\alpha V_k(\omega) + \beta V_{k'}(\omega) \geq (\sum_i 1/s_i)^{-1}$ with $s_i = \alpha/N^i_k + \beta/N^i_{k'}$, attained at $\omega^\star_i \propto N^i_k N^i_{k'}/(\alpha N^i_{k'} + \beta N^i_k)$. Harmonic is the $\alpha = \beta$ special case.*

Theorem 3 is unconditional in the counts – a property that distinguishes harmonic from the min-count variant, which has no such guarantee. Theorem 4 provides the oracle benchmark; in synthetic experiments substituting the true α, β into $\omega^\star$ yields variance within ±2% of harmonic. Complete proofs and additional sensitivity analyses are provided in the supplementary material.

Remark. The asymmetric analogue $\alpha V_k(\omega^h) + \beta V_{k'}(\omega^h) \leq \alpha V_k(\mathbf{1}) + \beta V_{k'}(\mathbf{1})$ does not hold unconditionally; a closed-form $n = 2$ counterexample requires strongly anti-correlated $(N_k, N_{k'})$ (each cell informative at exactly one position), a regime we have not seen in realistic count distributions. The $\omega^\star$ of Theorem 4 remains the exact oracle for all (α, β).

3 Plug-in Asymptotic Variance as Diagnostic

$\widetilde{V}$ in closed form. The delta method applied to $\hat{R}_{k,k'} = A/B$ gives

$$\widetilde{V}(\omega) = \frac{\sum_i \omega_i^2\, \theta^i_k(1-\theta^i_k)/N^i_k}{A^2} + \frac{\sum_i \omega_i^2\, \theta^i_{k'}(1-\theta^i_{k'})/N^i_{k'}}{B^2}, \tag{2}$$

Table 1: Plug-in SE vs. click-resample SE across KDD subsamples, and ARE(harmonic, uniform) at each scale. Click resample only computed at small f for runtime; agreement is within 1–2% where measured. ARE grows with data volume because heavy-tail mass becomes more pronounced at larger scales.

f	pairs	$\widehat{\mathrm{SE}}_U^{\mathrm{plug}}$	$\widehat{\mathrm{SE}}_U^{\mathrm{boot}}$	$\widehat{\mathrm{SE}}_H^{\mathrm{plug}}$	ARE
10^{-3}	4.3K	0.131	0.132	0.111	1.66
10^{-2}	54K	0.039	0.038	0.028	2.21
10^{-1}	542K	0.012	–	0.007	3.07
1.0	4.61M	0.0042	–	0.0020	**4.53**

where $A = \sum_i \omega_i \theta_k^i$, $B = \sum_i \omega_i \theta_{k'}^i$, and clicks at the two positions are conditionally independent given counts. The standard error of $\hat{R}_{k,k'}$ is then $\mathrm{SE}(\hat{R}_{k,k'}) = |\hat{R}_{k,k'}|\sqrt{\widetilde{V}(\omega)}$. Plugging in shrinkage estimates $\hat{\theta}_k^i = (C_k^i + \bar{\theta}_k)/(N_k^i + 1)$ with $\bar{\theta}_k$ the marginal position-k CTR yields the textbook self-normalised IS variance estimator [9–11].

ARE.. For schemes a, b, $\mathrm{ARE}(a, b) = \widetilde{V}(\omega_b)/\widetilde{V}(\omega_a)$: scheme b needs $\mathrm{ARE}(a, b)$ times the data of a for equal precision. $\widetilde{V}$ scales as $1/\mathrm{ESS}(\omega)$, with Kong's effective sample size [9] $\mathrm{ESS}(\omega) = (\sum_i \omega_i)^2/\sum_i \omega_i^2$.

Why pair-bootstrap is misaligned here. A naive bootstrap that resamples (q, d) pairs estimates a different quantity than the conditional-on-counts efficiency we care about, for three reasons. *(i)* Averaging over $\sim 60\%$ near-singleton pairs under uniform produces a smooth resampling distribution whose tightness reflects averaging-of-noise rather than estimator precision. *(ii)* Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, so concentrated weight schemes are systematically penalised by the resampling design, not by their statistical behaviour. *(iii)* Pair-bootstrap entangles estimator efficiency with model fit; $\widetilde{V}$ separates them (Section 4). The appropriate empirical baseline against which to validate the plug-in is the click-resample (parametric Bernoulli bootstrap conditional on counts), which we adopt for calibration.

Calibration and stability at scale. Table 1 reports plug-in SE versus empirical click-resample SE across KDD sample fractions, together with ARE(harmonic, uniform) at each scale. Because both $\widetilde{V}$ and click-resample SE share the conditional Bernoulli + cross-cell-independence assumption, their agreement (1–2%) is best read as an implementation-correctness check rather than as model validation. The ARE *grows* with sample size ($1.66 \longrightarrow 4.53$) because the heavy-tailed count structure becomes more pronounced as more data is included. A query-cluster bootstrap on the same data reflects the heavy-tailedness of the *query* distribution (a small set of head queries holds most of the impressions) and is reported in supplementary material; it is a diagnostic of the underlying log, not of the estimator at fixed counts.

4 Real-Log Results: KDD Cup 2012 Track 2

Data. Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique $(q, \mathrm{ad}, \mathrm{pos})$ cells, 8.2M clicks (CTR 3.49%). Pair $(1, 2)$ interventional set: 4.61M (q, ad) pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \leq 5$.

Table 2: Plug-in delta-method asymptotic variance $\widetilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\text{pairs}} = 4.61$M). Harmonic, min, and the count-only oracle all reach ARE ≈ 4.5. The full-θ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
Min	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	**4.53**	1,849
Count-oracle (0.5,1)	1.715	2.02e-3	4.57	1,760
θ-aware oracle	2.343	0.148	0.002	1.1

Table 3: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ tertile on full KDD (the bottom quartile boundary collapses into the lowest tertile because singletons dominate). Schemes are tied on the noise-dominated strata; harmonic dominates $\sim 5\times$ on the high-information stratum where the actual estimation signal lives.

$N_1 + N_2$ tertile	n pairs	$\widetilde{V}_U$	ARE(h, u)
Lowest	1.99M	2.3e-5	1.02
Middle-low	1.38M	1.9e-5	1.09
Highest	1.25M	7.6e-6	**4.98**

Headline (Table 2). ARE(harmonic, uniform) = 4.53: **uniform requires** $\sim 4.5\times$ **the impressions of harmonic for matching precision** on the relevance-agnostic surrogate. The min-count proposal achieves 4.41 on the same surrogate (−2.7% relative): harmonic and min are essentially tied empirically on KDD, and we recommend harmonic because it carries the unconditional guarantee (Theorem 3) and the parameter-free derivation, not because of a large empirical margin. The count-only oracle (4.57) is the ceiling within the surrogate. The θ-aware oracle, which would be optimal under classical inverse-variance theory with known r_i, plugs in $\hat{r}_i$ that is too noisy on the singleton-dominated tail and collapses to ARE = 0.002: a substantive finding rather than a side note—it shows that count-only weighting is robust precisely where plug-in IVW fails.

Adjacent pair (2,3) and regularised θ-oracle. At pair $(2, 3)$ (2.1M pairs), plug-in ARE is 3.14 and $\hat{R}_{2,3} = 1.73$ (vs. 1.71 at $(1, 2)$): the gain shrinks as imbalance decreases lower in the ranking, and $\hat{R}$ is consistent across pairs. Beta-shrinkage of $\hat{\theta}_i$ at $a \in \{1, \ldots, 1000\}$ pseudo-counts lifts the θ-oracle's ARE only from 0.002 to 0.61 – the count-only surrogate is the right object to optimise at this scale.

Asymmetric surrogate (empirical). With $\hat{p}_1 = 0.0501$, $\hat{p}_2 = 0.0256$ estimated from the marginal CTRs, $\alpha = 18.97$, $\beta = 38.08$, $\alpha/\beta \approx 0.50$. Harmonic reduces $\alpha V_k + \beta V_{k'}$ by a factor of 4.75 versus uniform. Both this number and the symmetric reduction of 4.84 are deterministic functionals of the observed counts—the gap reflects the intrinsic asymmetry of (α, β), not Monte-Carlo noise—and the two

Table 4: Count-only weighting schemes compared on full KDD via plug-in $\widetilde{V}$. Schemes that concentrate more aggressively give lower $\widetilde{V}$ but smaller ESS. Harmonic is the inverse-variance optimum among count-only schemes (Theorem 2).

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
$\log(1 + \min(N_k, N_{k'}))$	1.651	3.18e-3	1.71	3.17M
$\sqrt{N_k N_{k'}}$	1.706	2.28e-3	3.55	1,246
$\min(N_k, N_{k'})$	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	**4.53**	1,849
Harmonic, cap@99th pct	1.606	2.40e-3	2.83	765K

are within 2% of each other. The closed-form asymmetric counterexample of Section 2 requires strongly *anti-correlated* $(N_k, N_{k'})$ across cells (each cell informative at exactly one position); on KDD the per-pair counts are positively correlated, as items popular at position 1 are typically also popular at position 2, and we conjecture this is generic for production logs.

Stratified ARE (Table 3). Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by 5× on the high-information stratum. The gain comes from *not* treating singletons as equal to million-impression cells.

Weight-cap robustness. The harmonic ESS is 1,849 out of 4.61M pairs: concentration on a small subset of high-count cells is what produces the surrogate $\widetilde{V}$ reduction, and the trade-off is sensitivity to those cells, which we probe directly by capping. Capping ω_i^h at the 99.99th percentile (threshold 2,079; 462 pairs capped) lifts ESS to 5.2×10^4 and still gives ARE = 4.12; at the 99th percentile, ARE = 2.83 (ESS 7.6×10^5); even at the 90th, ARE = 1.86. Harmonic beats uniform robustly across capping levels.

Comparison of count-only weighting schemes. Table 4 compares harmonic to the natural alternatives on full KDD. Among count-only weights, harmonic is best on $\widetilde{V}$. Log-counting is too flat (treats singletons nearly as well as million-impression cells); the geometric mean and min-count concentrate more aggressively but neither matches the inverse-variance optimum. Capping at the 99th percentile preserves a 2.83× advantage but shifts the estimand ($\hat{R}$: 1.71 → 1.61), consistent with the estimand discussion below.

Estimand under PBM mis-specification. PBM is a known approximation. The population limit of $\hat{R}$ under weight ω is $R_\omega = \sum_i \omega_i \theta_k^i / \sum_i \omega_i \theta_{k'}^i$, which equals $p_k/p_{k'}$ only when $\theta_k^i/\theta_{k'}^i$ is constant across cells (PBM holds exactly). Under mis-specification each scheme targets a different weighted average of cell-specific ratios (uniform → 1.68, harmonic → 1.71, θ-aware oracle → 2.34 on KDD). The plug-in $\widetilde{V}$ comparison is a precision statement at fixed ω and does *not* claim that one R_ω is the "correct" position-bias ratio; choosing ω is a precision/estimand trade-off.

Quantifying PBM mis-specification. Cochran's Q on the 16,132 high-information pairs (≥ 5 clicks at each position) gives $Q/\text{df} = 3.83$: per-cell odds-ratio variability is ∼ 3.8× binomial alone. This is the quantitative driver of the estimand drift above; $\widetilde{V}$ keeps the efficiency comparison well-posed by working at fixed ω.

Table 5: MSE decomposition under three explicit PBM violations (anchor imbalance 50:1, split 80/20, 5,000 trials). Count-only schemes share comparable bias^2 within each model; harmonic achieves the lowest variance, hence the lowest MSE.

Model	Scheme	MSE	Variance	Bias^2
Trust	Uniform	0.00487	0.00486	1.0e-5
	Min	0.00298	0.00297	2.0e-5
	Harmonic	**0.00272**	0.00272	0.0
Cascade	Uniform	0.01376	0.00276	0.01100
	Min	0.01242	0.00171	0.01071
	Harmonic	**0.01252**	0.00147	0.01105
Pos.-dep.	Uniform	0.04086	0.03973	0.00113
	Min	0.03896	0.03770	0.00127
	Harmonic	**0.03724**	0.03613	0.00111

5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

Anchor-item imbalance sweep (synthetic). Following [1]: M=10 positions, $p_k = k^{-\eta}$ with $\eta = 1$, AdjacentChain estimator. A fraction s of documents are "anchors" with heavy traffic at one position ($N \sim \text{Unif}(50\rho, 200\rho)$) and light traffic at the other ($N \sim \text{Unif}(1, 51)$); the remaining $1 - s$ are balanced ($N \sim \text{Unif}(25, 75)$ at both positions). Here s is the traffic split and ρ is the imbalance ratio (disjoint from r_i above); $s \in \{0.5, \ldots, 0.95\}$, $\rho \in \{1, \ldots, 100\}$, 500 trials per cell. At $s = 0.80$, $\rho = 100$ harmonic reduces variance by 43.6% vs. uniform and 5% vs. min; reduction grows to ∼ 54% at $s = 0.90, \rho = 50$. Across the full (s, ρ) grid the data-equivalence factor $\text{DEF}(s, \rho) = b_{\text{uni}}(s, \rho)/b_{\text{harm}}(s, \rho)$ – the ratio of base impressions per document at which each scheme first hits a fixed MSE threshold – has median 1.31, mean 1.30, max 1.94. Highly imbalanced cells achieve up to 2× data savings; near-balanced cells ($\rho = 1$) saturate at DEF ≈ 1.

Bias comparability under PBM violations. We additionally simulate three explicit PBM violation models – trust bias, cascade, and position-dependent relevance [1, 5] – at imbalance ratio 50:1, split 80/20, 5,000 trials. Across all three models the count-only schemes (uniform, min, harmonic) exhibit *comparable* bias^2, so harmonic's lower variance translates directly into lower MSE (Table 5). Under cascade, where history-dependent examination introduces a substantial structural mismatch (bias^2 ≈ 0.011 for every ratio-unbiased scheme), the variance edge is preserved but the practical MSE gain shrinks because bias dominates.

Yahoo-LTR semi-synthetic. Same simulator with graded relevance from the Yahoo Learning to Rank Challenge Set 1 test split (6,936 queries; relevance grades $g \in \{0, \ldots, 4\}$ mapped to $r(q, d) = (2^g - 1)/15$ following NDCG; click model $P(C = 1 \mid q, d, k) = p_k\, r(q, d)$; 200 trials). Harmonic reduces variance by ∼ 40% vs. uniform at imbalance $\rho = 50$ (Var $1.5 \times 10^{-4} \rightarrow 8.7 \times 10^{-5}$). With 200 trials, a delta-method approximation places the 95% CI on the variance-ratio at $[0.49, 0.71]$, so the gain is well separated from 1.0 but not tightly pinned. Aggressive clipping trades variance for bias and yields ∼458× higher MSE.

6 Related Work

Intervention harvesting was introduced by [1]; [13] jointly estimates relevance and propensities via EM. Self-normalised weighted ratio estimators are standard in importance sampling and survey statistics [3, 9, 10]; inverse-variance weighting is canonical in meta-analysis [4]. The closest off-policy analogue is SNIPS [2, 11]. PBM violations on real logs are documented in [6, 14].

Relation to Agarwal et al.: set construction vs. weighting. The framework of [1] has two separable ingredients: (i) *interventional-set construction* – which position pairs to harvest and how to chain them (AllPairs over all (k, k'), AdjacentChain through adjacent pairs) – and (ii) the *per-cell weighting* inside the ratio estimator of Eq. (1), which is uniform ($\omega_i \equiv 1$). Our contribution is confined to (ii): harmonic is a drop-in replacement for the weight in $\hat{R}_{k,k'}$ and leaves set construction untouched, so it composes with AllPairs, AdjacentChain, or any other harvesting set. This is why uniform weighting – not AllPairs – is the fair baseline: uniform is the only weighting published for these estimators, and AllPairs and AdjacentChain both inherit it, so isolating the weighting effect requires holding the set fixed and varying ω alone. Our KDD results (Section 4) fix the adjacent-pair set of [1] and vary only ω; the 4.5× efficiency gain is therefore attributable to weighting, not to a different set-construction choice.

Comparison to PBM-EM and outlook. Vanilla no-feature PBM-MLE (one r_i per cell) on KDD $(1, 2)$ collapses to $\hat{R}_{1,2} = 1.682$ – the uniform estimand – because the model is over-parameterised on singleton-dominated data. Wang et al.'s feature-based EM [13] avoids this but needs features unavailable in the anonymised aggregation; on synthetic PBM data, the same MLE recovers $R = 2.0$ with SE $= 0.035$ (harmonic 0.040, uniform 0.055, 500 trials), confirming implementation. *In short:* harmonic is the count-only surrogate-IVW optimum and reduces conditional click-noise variance by 4.5× vs. uniform on KDD with our two proposals empirically tied; harmonic is preferred for the unconditional guarantee of Theorem 3. *Scope.* We target propensity-estimation efficiency at a fixed estimand rather than end-to-end ranking. Under IPS, ranking quality is governed by the propensity *shape* – recovered by any consistent harvesting estimator – whereas estimation efficiency governs how much log data is needed to pin that shape down; we therefore do not claim a direct improvement in ranking metrics such as nDCG. Future work: feature-based EM comparison; click models beyond PBM [6, 14].

GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 2–4 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review. Full code with unit tests and the data-processing pipelines needed to reproduce every result are publicly available at https://github.com/alemagnani/surrogate-ivw-intervention-harvesting.

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